A CARTESIAN SQUARE OF FINITE TYPE ALGEBRAS WITH SURJECTIONS KEEPS THE FOURTH CORNER OF FINITE TYPE

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The fibre product of two finite type algebras over a Noetherian ring with surjections is of finite type. the statement counted by the sibling card S1-algebra-fibre-product-finite-type at 326-362, cited at proof line 380. Is the case to which the induction step reduces.
- Fibre products over a finite index set and the identification of the last square as an instance of the two-factor statement. used at proof lines 382-394. Organises the induction and produces the square to which the cited statement applies.

Source reference. Original lemma 08NI: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10991–10991. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 00IT: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 691–691. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.2. *Let R be a Noetherian ring. Let I be a finite set. Suppose given a cartesian diagram*

$$\begin{array}{ccc} \prod B_i & \xleftarrow{\prod \varphi_i} & \prod A_i \\ {\scriptstyle \prod \psi_i}\uparrow & & \uparrow \\ Q & \longleftarrow & P \end{array}$$

with ψ_i and φ_i surjective, and Q, A_i, B_i of finite type over R. Then P is of finite type over R.

Proof. Follows from Lemma 00IT and induction on the size of I. Namely, let $I = I' \amalg \{i_0\}$. Let P' be the ring defined by the diagram of the lemma using I'. Then P' is of finite type by induction hypothesis. Finally, P sits in a fibre product diagram

$$\begin{array}{ccc} B_{i_0} & \longleftarrow & A_{i_0} \\ \uparrow & & \uparrow \\ P' & \longleftarrow & P \end{array}$$

to which the lemma applies. □